

response curve (AUC) was then calculated in order to obtain a single number, which indicates the total increase in blood glucose during the two-hour test period in that subject as a result of ingesting that food. A glycaemic index (GI) value for each test sugar was then calculated for each subject by dividing their two-hour blood glucose AUC value for the test food by their average two-hour blood glucose AUC value for the reference food and multiplying by 100 to obtain a percentage score.

$$\text{GI value (\%)} = \frac{\text{Blood glucose AUC value for the test food}}{\text{Average AUC value for the equal carbohydrate portion of the reference food}} \times 100$$

Average AUC value for the equal carbohydrate portion of the reference food

Due to differences in body weight and metabolism, blood glucose responses to the same food or drink can vary between different people. The use of the reference food to calculate GI values reduces the variation between the subjects' blood glucose results to the same food arising from these natural differences. Therefore, the GI value for the same food varies less between the subjects than their glucose AUC values for this food.

Table 2 - GI for samples prepared in example 1

Sample	GI ($\pm$ SEM)	GI category
Reference	100 $\pm$ 0	High GI
Control sugar	68 $\pm$ 3	Medium GI
Sugar 1	53 $\pm$ 4	Low GI
Sugar 2	70 $\pm$ 4	High GI

Example 5 - Relationship between GI and polyphenol, glucose, fructose and moisture content

Increasing glucose and fructose in low GI sugar can affect the GI of sugar. In many unrefined sugars, as sucrose content decreases reducing sugar content increases. The increase in reducing sugars can increase the GI of the unrefined sugar. This effect is counterintuitive and unexpected. Most consumers understand that less refined products are healthier or better for you. However, that is not necessarily the case for unrefined sugar. The healthiest sugars minimise reducing sugar content without refining out all the polyphenols responsible for the low GI. There is a "sweet spot" in the extent to which sugar is refined where GI remains low. The inventors of the present invention have researched less refined sugars by varying the extent of the massecuite washing. Too much washing removed the majority of polyphenol content and increased the GI. Too